Phase 3 Freeze Report

AI Intelligence Architecture

DeepMindQ CRM -- Enterprise AI Governance & Intelligence Foundation

July 19, 2026

1. Final Architecture Diagram

The DeepMindQ CRM Phase 3 architecture is a layered system with clear separation between the user-facing frontend, API routing layer, AI governance middleware, LLM primitives, research intelligence engine, and persistent storage. Each layer communicates through well-defined contracts, and all AI calls are forced through the governance gate.

Layer Stack

- **Frontend:** Next.js + Tailwind CSS + shadcn/ui component library. Server-side rendering with React Server Components for data-heavy views.
- **API Layer:** Next.js Route Handlers under `src/app/api/`, grouped by domain: `g-ai`, `g-crm`, `g-data`, `g-strategy`, `g-outreach`, `g-system`.
- **AI Governance Layer:** `ai-governance.ts` exports `governedAICall()` (company-scoped, 6 checks) and `governedAICallAggregate()` (non-company, hallucination prevention).
- **LLM Primitives:** `zai-helpers.ts` provides `callLLM()` and `webSearch()`. Direct import by API routes is forbidden; all calls go through governance.
- **Research Engine:** 6-step pipeline -- search, evidence storage, LLM extraction with governance, field validation, per-field confidence scoring, intelligence storage.
- **Intelligence Contract:** `intelligence-contract.ts` implements freshness calculation (4 domain half-lives), confidence aggregation, and context building for AI prompts.
- **Database:** PostgreSQL (Neon) via Prisma ORM. Schema includes company, contact, signal, evidence, capability, and audit tables.

Layer	Module	Responsibility
Frontend	Next.js + Tailwind + shadcn/ui	User interface, server components
API	src/app/api/g-*	Domain-grouped route handlers
Governance	ai-governance.ts	governedAICall / governedAICallAggregate
LLM	zai-helpers.ts	callLLM, webSearch primitives
Research	research-engine/	6-step intelligence pipeline
Intelligence	intelligence-contract.ts	Freshness, confidence, context

Database	PostgreSQL + Prisma	Persistent storage, ORM

2. Complete Data Flow

The intelligence chain describes the end-to-end journey from company creation through AI-generated deliverables. Every step is tracked, governed, and auditable.

- **Step 1 -- Company Created via CRM:** User adds a company through the CRM interface. A database record is created and the company becomes eligible for research.
- **Step 2 -- Research Triggered (6-Step Pipeline):** Four parallel Tavily web searches execute (general news, funding, hiring, tech changes). Results flow through: evidence storage, LLM extraction with governance, field validation, per-field confidence scoring, and final intelligence storage.
- **Step 3 -- Evidence Collected:** Per-field evidence records are created, each tracking source URL, content snippet, and extraction timestamp. Multiple sources per field enable corroboration scoring.
- **Step 4 -- Confidence Scored Per Field:** Each intelligence field receives a confidence score (0.0 to 1.0) based on three factors: source quality, recency, and corroboration across multiple sources.
- **Step 5 -- Freshness Calculated (4 Domains):** Exponential decay model with domain-specific half-lives: profile (90 days), signals (14 days), contacts (45 days), technology (60 days). Each domain is independently scored and averaged.
- **Step 6 -- Signals Detected:** `governedAICallAggregate()` analyzes evidence for 10 signal types (funding rounds, hiring sprees, product launches, etc.). Structured extraction produces typed, scored signals.
- **Step 7 -- RFP/RFI Fields Populated:** When signals indicate procurement activity, structured fields are populated: opportunityType, deadline, buyingArea, techRequirement, serviceRequirement, matchingCapability.
- **Step 8 -- Capability Matching:** SignalCapabilityMatch table links signals to capabilities using weighted scoring: category (30%), keyword (30%), business problem (20%), impact (5%), remaining 15% from reasoning quality.
- **Step 9 -- AI Outputs Generated:** Through governance gates, the system produces account briefs, email drafts, and conversation plans. Each generation creates an AIGenerationAudit record.
- **Step 10 -- Audit Trail:** Every AI generation creates an AIGenerationAudit record with 15 fields including timestamps, governance results, model used, confidence/freshness at time of generation, and the full prompt sent.

3. AI Governance Architecture

AI governance is the central safety mechanism. Two gate functions enforce quality thresholds before any LLM call is permitted. This prevents hallucination, stale data usage, and low-confidence outputs from reaching users.

3.1 Gate Functions

- **governedAICall():** Company-specific gate. Requires a companyId and performs 6 governance checks before allowing the LLM call. Used for all company-scoped AI generation (briefs, emails, plans, insights).
- **governedAICallAggregate():** Non-company gate. Focuses on hallucination prevention. Used for cross-company analysis, command-center queries, data health, and research extraction where no single company context applies.

3.2 Six Governance Checks

Check	Description	Failure Action
research_exists	Company has completed research cycle	Block: require research
research_confidence	Overall confidence meets type threshold	Block: insufficient data
freshness_score	Composite freshness meets type threshold	Block: data too stale
staleness	No single domain exceeds max staleness days	Warn: partial degradation
capability_match	Capability score available for RFP types	Warn: no match data
recent_intelligence	Intelligence updated within acceptable window	Block: trigger re-research

3.3 Per-Type Thresholds

Generation Type	Min Confidence	Min Freshness	Max Staleness (days)
email_draft	0.60	25	60
account_brief	0.50	20	90
conversation_plan	0.55	20	60
signals	0.40	15	30
insights	0.45	20	60
recommendations	0.50	20	60
summarize	0.30	10	120
enrich	0.35	15	90

3.4 Hallucination Prevention

15 hallucination prevention rules are injected into every LLM prompt. These rules enforce:

- Never fabricate data not present in the provided context

- Explicitly state "unknown" when information is missing
- Distinguish between confirmed facts and inferences
- Cite source URLs when available
- Flag low-confidence claims with uncertainty markers
- Never extrapolate financial figures or headcount estimates
- Never invent product names, feature lists, or partnership details
- Always ground analysis in provided evidence only

Staleness-based confidence modifier: For each stale domain, confidence is reduced by 5% to 15% depending on how far past the half-life the data has aged. This modifier is applied at generation time, after the base confidence check passes.

Build-time guard: `scripts/check-governance.sh` runs 5 automated checks to ensure no API route imports `callLLM` directly from `zai-helpers.ts`. This is integrated into `npm run lint` as a pre-commit hook.

4. All AI Entry Points

The following table catalogs every AI generation site in the codebase. All 40 sites use one of the two governance gate functions -- there are zero direct `callLLM()` calls from API routes.

4.1 governedAICall Sites (23)

#	Site	Context
1	signals	Signal detection for a company
2	opportunities	Opportunity analysis per company
3	insights	Company insight generation
4	suggested_contacts	Contact recommendation
5	recommendations	Strategic recommendations
6	conversation_plan	Meeting conversation plan
7	enrich	Company data enrichment
8	generate-pdf (call 1)	PDF executive summary
9	generate-pdf (call 2)	PDF detailed analysis
10	generate-pdf (call 3)	PDF recommendations
11	account-brief	Account brief generation
12	summarize (company)	Company summary

13	summarize (contact)	Contact summary
14	generate-ppt	Presentation generation
15	command-center query	Command center AI query
16	email draft	Outreach email drafting
17	ab-test	A/B test variant generation
18	strategy	Strategic analysis
19-23	workflow email gen (x5)	Workflow-triggered email generation

4.2 governedAICallAggregate Sites (17)

#	Site	Context
1	query_parsing	Natural language query parsing
2	knowledge_enrichment	Cross-company knowledge enrichment
3	research_agent_person	Person research agent
4	relationship_memory (1/5)	Relationship context building
5	relationship_memory (2/5)	Relationship strength analysis
6	relationship_memory (3/5)	Interaction history synthesis
7	relationship_memory (4/5)	Stakeholder mapping
8	relationship_memory (5/5)	Relationship recommendation
9	command-center query	Aggregate dashboard query
10	data_health_analysis (1/3)	Data completeness check
11	data_health_analysis (2/3)	Data quality scoring
12	data_health_analysis (3/3)	Data freshness assessment
13	playbook_generation	Sales playbook creation
14	research_extraction (1/2)	Research field extraction
15	research_extraction (2/2)	Research validation pass
16	signal_detection	Cross-company signal scan
17	generate-email (aggregate)	Template-based email generation

5. Database Schema Changes

Phase 3 introduces significant schema additions to support AI governance, evidence tracking, RFP/RFI intelligence, and capability matching.

5.1 CompanySignal -- 7 New RFP/RFI Fields

Field	Type	Purpose
opportunityType	String?	RFP, RFI, RFT, or contract type
publicationDate	DateTime?	When the opportunity was published
deadline	DateTime?	Submission deadline
buyingArea	String?	Procurement category / buying organization
techRequirement	String?	Required technology capabilities
serviceRequirement	String?	Required service deliverables
matchingCapability	String?	Best-matched internal capability

5.2 AIGenerationAudit -- New Table (15 fields, 7 indexes)

Tracks every AI generation for compliance, debugging, and quality analysis. Key fields include: id, companyId, generationType, inputContext (JSON), outputResult (JSON), promptSent (JSON), modelUsed, confidenceScore, freshnessScore, governanceResult (JSON), createdAt, processingTimeMs, userId, metadata (JSON). Seven indexes cover companyId, generationType, createdAt, and composite lookups.

5.3 Evidence -- New Table (7 indexes)

Per-field evidence tracking. Each record links a piece of evidence to a specific intelligence field on a company. Fields: id, companyId, fieldName, sourceUrl, content, extractedValue, confidence, extractedAt, researchBatchId. Seven indexes enable fast lookup by company, field, batch, and source.

5.4 SignalCapabilityMatch -- New Table (6 indexes)

Links detected signals to internal capabilities with reasoning. Fields: id, signalId, capabilityId, categoryScore, keywordScore, businessProblemScore, impactScore, totalScore, reasoning, createdAt. Six indexes cover signal, capability, and score-based queries.

5.5 CompanyResearchCard -- 5 New Intelligence Fields

Field	Type	Purpose
strategicPriorities	Json?	Extracted strategic priorities list
businessProblems	Json?	Identified business challenges

transformationAreas	Json?	Digital transformation focus areas
technologyThemes	Json?	Technology stack and themes
structuredTechLandscape	Json?	Structured technology landscape map

4 Freshness Timestamp Fields: profileLastResearched, signalsLastResearched, contactsLastResearched, technologyLastResearched -- each tracking when that intelligence domain was last updated.

6. API Contracts

Key API endpoints follow a consistent pattern: POST with JSON body, governed AI call in the handler, and structured JSON response. All AI endpoints return governance metadata alongside the generated content.

Endpoint	Method	Gate Function	Notes
/api/g-ai/ai__chat	POST	governedAICallAggregate	General AI chat, non-company-scoped
/api/g-ai/ai__account-brief	POST	governedAICall	Requires companyId, confidence >= 0.50
/api/g-ai/ai__conversation-plan	POST	governedAICall	Requires companyId, confidence >= 0.55
/api/g-crm/contacts/:id/generate-email	POST	governedAICall	Generates draft, requires approval
/api/g-outreach/drafts	POST	governedAICall	Human approval required before sending

Human-in-the-loop enforcement: Email drafts created via the outreach API are stored with status "draft" and cannot be sent without explicit user approval. The sequence processing system auto-approves drafts only when a cron job is configured (not active in current deployment).

7. Module Dependency Map

The dependency map enforces the architectural rule that API routes may only call `ai-governance.ts`, never `zai-helpers.ts` directly. This ensures every LLM call passes through governance.

Module	Depends On	Allowed Import
ai-governance.ts	zai-helpers.ts	callLLM only
research-engine/researcher.ts	ai-governance.ts, evidence.ts, signals.ts	Full governance gate
research-engine/signals.ts	ai-governance.ts	governedAICallAggregate
research-engine/evidence.ts	zai-helpers.ts	Types only (no LLM calls)
intelligence-contract.ts	Prisma client	Database read/write

email-generation.ts	ai-governance.ts, intelligence-contract.ts	Governed + context
All API routes (g-ai/*)	ai-governance.ts	NEVER zai-helpers.ts for LLM

Build-time enforcement: `scripts/check-governance.sh` scans all API route files for direct imports of `callLLM` from `zai-helpers.ts`. Any violation causes the lint step to fail, blocking the CI/CD pipeline.

8. Capability Intelligence Model

8.1 SignalCapabilityMatch Scoring

Factor	Weight	Method
Category Match	30%	Signal type mapped to capability category taxonomy
Keyword Match	30%	Signal text tokens matched against capability keywords
Business Problem	20%	Signal pain points aligned to capability problem space
Impact Score	5%	Estimated business impact of the match
Reasoning Quality	15%	LLM reasoning coherence and evidence citation

8.2 Signal Type Mappings

Signal Type	Detection Trigger	Typical Capability Match
funding_round	Investment announcement	Growth advisory, scaling services
hiring_spree	Rapid headcount growth	HR tech, onboarding solutions
product_launch	New product announcement	Marketing, distribution, support
leadership_change	C-suite appointment	Strategy consulting, integration
partnership	Partnership or alliance	Integration, co-development
acquisition	Acquisition activity	M&A; advisory, integration services
expansion	Geographic or market expansion	Localization, market entry
tech_migration	Technology stack change	Migration services, new platform
regulatory	Regulatory or compliance event	Compliance consulting, audit
contract_award	Government or large contract	Delivery, staffing, fulfillment

8.3 Composite Intelligence Score

The composite intelligence score aggregates multiple dimensions into a single 0-100 score used for company prioritization and tier assignment:

Dimension	Weight	Source
Data Completeness	25%	CompanyResearchCard field population ratio
Evidence Quality	20%	Average evidence confidence across all fields
Freshness	15%	Composite freshness from 4 domain half-lives
Signal Activity	20%	Recent signal count weighted by recency
Contact Coverage	10%	Number and quality of tracked contacts
Engagement History	10%	Email opens, meetings, interactions

8.4 Tier Thresholds

Tier	Score Range	Action
Hot	>= 70	Prioritize outreach, fast-track engagement
Warm	>= 40	Nurture with targeted content and monitoring
Cold	>= 15	Maintain passive monitoring, periodic re-research
Dormant	< 15	Archive or deprioritize; re-evaluate quarterly

9. Known Technical Debt

The following items are documented technical debt from Phase 3. Each has been evaluated and a deliberate decision made to defer resolution.

9.1 Sequence Processing Auto-Approval

`sequences__process.ts` auto-approves generated email drafts. No cron job is currently configured, so this code path is dormant. The design intentionally allows future automation when the team is ready to enable autonomous sending. **Risk:** If a cron is accidentally enabled without additional safeguards, drafts could be sent without human review.

9.2 modelUsed Field Default

The `modelUsed` field in AIGenerationAudit defaults to "governance-tracked" rather than recording the actual LLM model name (e.g., "gpt-4o", "claude-3.5-sonnet"). **Impact:** Audit records cannot be used to compare model performance. **Resolution:** Pass model name through the governance gate in Phase 4.

9.3 SignalCapabilityMatch Name Join Missing

`getSignalCapabilityMatches()` does not join signal names or capability names, returning empty strings for display fields. **Impact:** Frontend must perform additional lookups or display IDs. **Resolution:** Add Prisma include clauses for Signal and CapabilityAsset relations.

9.4 callLLM Export from zai-helpers.ts

`callLLM` remains exported from `zai-helpers.ts` because `ai-governance.ts` needs to call it internally. The build-time governance check ensures no API route imports it directly. **Risk:** Low -- the guard script catches violations at build time.

10. Phase 4/5/6 Dependency Rules

Phase 3 establishes the foundation that all future phases depend on. The following rules ensure backward compatibility and architectural integrity.

10.1 Phase 4 -- Opportunity Intelligence

The RFP/RFI fields on CompanySignal (opportunityType, deadline, buyingArea, techRequirement, serviceRequirement, matchingCapability) are the foundation for the opportunity pipeline. Phase 4 builds an opportunity engine that reads signals where `opportunityType IS NOT NULL` and creates structured opportunity records. **Rule:** Do not modify the signal detection prompt format -- Phase 4 queries depend on the current structured output schema.

10.2 Phase 5 -- Predictive Scoring

Uses the composite intelligence score from `intelligence-contract.ts` as the base signal. Phase 5 adds time-series tracking of signal-to-opportunity conversion rates and predicts future opportunity likelihood. **Rule:** The composite score formula (data 25%, evidence 20%, freshness 15%, signals 20%, contacts 10%, engagement 10%) must remain stable -- changes require a versioned schema migration.

10.3 Phase 6 -- Autonomous Recommendations

IRON RULE: Phase 6 MUST remain human-in-the-loop. AI recommends, human decides. The system must never auto-send emails, auto-approve drafts, or auto-enroll companies in sequences without explicit human approval. This is a non-negotiable architectural constraint that supersedes all optimization goals.

11. RFP/RFI Intelligence Foundation

Phase 3 lays the groundwork for automated RFP/RFI detection and matching. The current implementation uses web search as the primary intelligence source, with the architecture designed to incorporate additional sources in future phases.

11.1 Monitored Sources

Current (Tavily Web Search):

- General news and press releases
- Funding announcements (Crunchbase, PitchBook coverage)
- Hiring activity (LinkedIn, Indeed, company career pages)
- Technology changes (blog posts, stack share updates)
- Partnership and acquisition announcements

Future Sources (Phase 4+):

- Government procurement portals (SAM.gov, TED, Contracts Finder)
- Industry-specific bid databases
- State/local government procurement sites
- Private sector RFP aggregators

11.2 Detection Method

LLM-based signal analysis with a structured extraction prompt. When research evidence contains procurement-related language, the signal detection prompt extracts: opportunityType (RFP/RFI/RFT/contract), deadline, buyingArea, techRequirement, and serviceRequirement. Each extracted field is validated against the governance thresholds before storage.

11.3 Capability Matching

The SignalCapabilityMatch table links RFP-related signals to internal CapabilityAsset records. Each match includes: category score (30%), keyword score (30%), business problem score (20%), impact score (5%), and an LLM-generated reasoning field explaining why the capability is relevant. The total score is used to rank capabilities for user review.

11.4 Phase 4 Readiness

This RFP/RFI intelligence foundation is the direct prerequisite for the Phase 4 Opportunity Intelligence engine. Phase 4 will: (1) add dedicated RFP source connectors, (2) build an opportunity pipeline that converts detected RFP signals into trackable opportunity records, (3) implement deadline-based prioritization, and (4) provide automated capability-gap analysis when no strong match is found.